

Code-Layout, Readability and Reusability

Date	04 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the OptiCrop codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed	Remarks
1	Consistent Indentation	Uniform 4-space indentation used throughout Python code	Yes	PEP8 compliant
2	Proper File Structure	Separate files for Flask app, database, templates, static resources	Yes	Well organized
3	Meaningful Variable Names	Variables like <code>fullname</code> , <code>email</code> , <code>prediction</code> , <code>features</code> are self-explanatory	Yes	Easy to understand
4	Function Names	Functions such as <code>home()</code> , <code>register()</code> , <code>login()</code> , <code>dashboard()</code> , <code>predict_route()</code> , <code>logout()</code> clearly indicate their purpose	Yes	Descriptive naming
5	Code Comments	Comments used to separate application sections	Yes	Improves readability

6	Modular Design	Database operations separated into <code>database.py</code> ; application logic in <code>app.py</code>	Yes	Reusable structure
7	No Redundant Code	Minimal duplicate code with reusable functions	Yes	Maintainable
8	Error Handling	<code>try-except</code> used in crop prediction	Yes	Basic exception handling

Reusable Components / Modules:

S.No	Component	Technology	Reused In	Reusability
1	Database Module (<code>database.py</code>)	Python, SQLite	User Registration, Login, Prediction History	High
2	Machine Learning Model (<code>crop_model.pkl</code>)	Scikit-learn	Crop Prediction	High
3	Flask Application (<code>app.py</code>)	Flask	Web Application Backend	High
4	Prediction Function	NumPy, Pickle	Crop Recommendation	High
5	HTML Templates	HTML, Bootstrap	User Interface	Medium

Overall Code Quality Assessment:

Aspect	Rating	Comments
Code Layout & Structure	5/5	Well-structured with separate backend, database, and templates
Readability	5/5	Clear variable names and organized functions
Reusability	5/5	Database and prediction logic can be reused in future versions
Documentation	4/5	Section comments present; adding a README would improve documentation
Overall Score	4.8/5	Well-organized Flask-based machine learning application